Optimizing dimensionality reduction in gene expression using autoresearch

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Abstract

High-dimensional gene expression data are noisy and difficult to model, motivating dimensionality reduction methods that produce compact representations for downstream analysis. In this project, I apply autoresearch—an agentic workflow for autonomous experimentation—to compute a low-dimensional embedding of gene expression data from the human brain and predict the chromosome class for each gene probe. Candidate pipelines were explored and selected using a three-metric scheme: silhouette score and precision@10 to assess global and local embedding quality, and balanced accuracy to assess classification performance. Autoresearch revealed Independent Component Analysis (ICA) as the most effective dimensionality reduction method, and later improved classification by using balanced class weights in Logistic Regression (LR). On the held-out test set, the final model achieved small but consistent gains over both a null model (no dimension reduction) and a PCA baseline with no evidence of overfitting.

1 Introduction

Autoresearch [1] is a recent development in machine learning in which an agentic AI system improves model performance through repeated, autonomous experimentation. In a traditional workflow, a human researcher proposes modeling changes, runs experiments, and interprets results; autoresearch automates these steps by having an AI agent iteratively modify a pipeline, evaluate each modification under a fixed scheme, and retain only changes that improve performance. As a result, autoresearch can be applied broadly to supervised learning problems with well-defined success metrics.

In this project, I apply autoresearch to optimize dimensionality reduction for gene expression data. It is often challenging to isolate true biological signal from noise in high-dimensional measurements of gene expression. Dimensionality reduction can help by compressing the data into a smaller set of components that preserve key sources of information. Moreover, inspecting the features that comprise each reduced component can provide insight into gene-level relationships that may underlie higher-level behaviors. For example, Sastry et al. [2] argue that the *E. coli* transcriptome largely consists of independently regulated modules by decomposing signals into approximately independent components. My work evaluates whether autoresearch can efficiently discover dimensionality reduction choices that improve both embedding quality and downstream classification performance.

2 Methods

All analyses were implemented in Python, and development was assisted by the GitHub Copilot AI Agent in VSCode. Work for this project was split into two parts: manual setup and autonomous experimentation via autoresearch.

2.1 Manual Setup

Data for this project were sourced from the Allen Brain Atlas [3]. This microarray dataset contains measurements for 58,692 gene probes taken across 946 regions in the human brain. Each probe is associated with 1 of 24 chromosomes in the human genome. After removing probes with missing labels, chromosomes were recoded into 5 groups by gene density to address class imbalance in the target variable, producing a more suitable distribution for classification (Figure 1). A testing set was saved for final model analysis by splitting 20% of the data using stratified sampling, and the remaining data were split into training (80%) and validation (20%) sets for each model run.

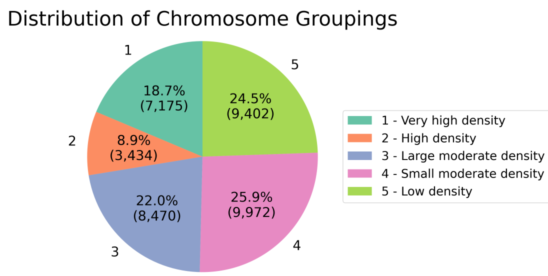

Figure 1: Distribution of the target variable after grouping chromosome labels by gene density. The distribution remains moderately imbalanced but is better suited for predictive modeling.

Additional preparation was necessary before allowing autonomous agent experimentation. The baseline model was defined using reasonable methods that performed well under simple manual exploration, selecting Principal Component Analysis (PCA) for dimensionality reduction and Logistic Regression (LR) for classification. Agent instructions in `program.md` and documentation in the main code files were manually specified to reflect the frozen and editable module structure.

2.2 Autonomous Stage

The autonomous stage used a three-metric evaluation scheme to assess embedding quality and downstream predictive performance. Silhouette score evaluated the global separability of class labels in the low-dimensional representation of gene expression, and precision@k (with $k = 10$) assessed local neighborhood structure in the embedding. After training the classification model using the low-dimensional embeddings, balanced accuracy assessed predictive performance while accounting for class imbalances. To avoid retaining changes driven by small validation fluctuations, an experiment was deemed successful only if it exceeded an improvement threshold (e.g. silhouette score increased by at least 0.005, smaller positive gains in both silhouette score and precision@10, or unchanged embeddings with a meaningful improvement in balanced accuracy).

Starting from the baseline, the AI agent ran initial autoresearch loops with an exploratory goal. While the agent was required to follow the specifications in `program.md`, it was otherwise permitted to explore a wide range of dimensionality reduction and classification methods under a fixed runtime budget. Results from this phase were used diagnostically to refine the agent instructions and, after early stagnation under the original balanced accuracy metric, to motivate the eventual three-metric evaluation scheme.

After this broad exploration stage, the autoresearch scope was narrowed to focus on the most notable gains. Additional restrictions constrained the agent to a smaller class of experiments (such as tuning hyperparameters with a fixed dimensionality reduction method and classifier). As evidence accumulated about unproductive directions, the agent also self-selected toward smaller, more targeted modifications rather than wholesale pipeline changes. The focused loop continued until the agent exhausted all reasonable proposals and failed to produce further improvements.

3 Results

A total of 58 autonomous experiments were completed over 4 autoresearch runs under the three-metric evaluation scheme. These included 2 baseline experiments, 3 kept experiments, 42 discarded experiments, and 11 failed experiments. Among the failures, the most common cause was exceeding the runtime limit, particularly early in development before the time allocation was increased from 3 minutes to 10 minutes per model. The best model from autoresearch used Independent Component Analysis (ICA; FastICA implementation) for dimensionality reduction and LR with balanced class weights for classification.

Figure 2 shows the evolution of autoresearch performance across all experiments. The most prominent improvement occurs early, when the agent discovers FastICA after only a few trials. This leads to an increase in silhouette score (from approximately -0.030 to -0.008) and precision@10 (from approximately 0.235 to 0.241), after which both embedding metrics plateau. In the middle of the trajectory, the agent explores a wide range of alternatives that produce highly variable results; occasional single-metric improvements are not retained because they fail to meet the improvement threshold or degrade companion metrics. As the autoresearch scope narrows toward model tuning, the agent makes one final improvement by introducing balanced class weights in the LR classifier. The dimensionality reduction model and embedding metrics remain unchanged, but this produces a modest improvement in classification performance.

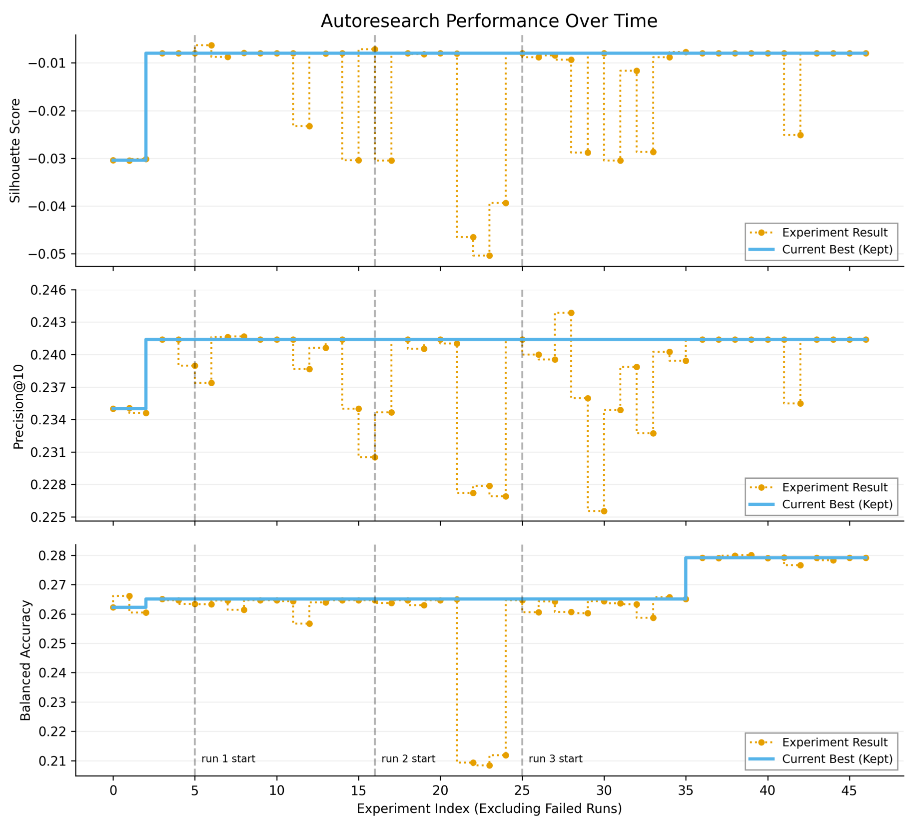

Figure 2: Progression of evaluation metrics across autoresearch experiments. The best kept silhouette score and precision@10 are achieved early after discovering FastICA, while a later improvement in balanced accuracy is obtained by tuning the classifier.

Assessment of the final model was completed on the reserved testing set (held out from all model development steps), providing a valid benchmark for performance on unseen data. The best model from autoresearch was compared with the null model (LR on the original gene expression data) and baseline model (PCA + LR). Figure 3 and Table 1 show small but noticeable improvements for the final model over both the null and baseline, with the clearest gain in silhouette score. Interestingly, the baseline model exhibits better classification performance than the null despite worse embedding metrics, suggesting that dimensionality reduction can benefit downstream prediction even when it does not improve label separability in the reduced space. The test set metrics are also comparable to the validation performance of the selected model, giving no indication of overfitting.

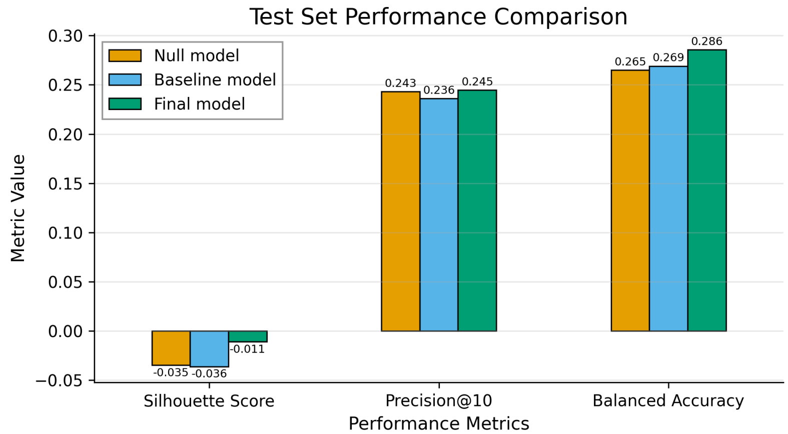

Figure 3: Test set performance metrics for the null, baseline, and final models. The final model attains a small but noticeable improvement over the null and baseline models across all three metrics.

Table 1: Test set performance metrics for the null, baseline, and final models.

Model	Silhouette Score	Precision@10	Balanced Accuracy	Train Time (s)
Null (No DR)	-0.035	0.243	0.265	12
Baseline (PCA)	-0.036	0.236	0.269	14
Final (FastICA)	-0.011	0.245	0.286	100

4 Discussion

Although autoresearch produced measurable gains, overall predictive performance remains modest (test balanced accuracy ≈ 0.286). This indicates that the chromosome class of a gene probe is difficult to predict from its expression profile alone, and that much of the variance in the expression data may not be informative for this task. Given the noisy, high-dimensional setting and the absence of domain-guided feature engineering, limited improvement from automated tuning is not surprising.

A key technical takeaway is that many plausible dimensionality reduction approaches did not improve performance. In particular, nonlinear methods such as UMAP and autoencoder-based embeddings were hypothesized to yield better representations due to their flexibility, but they consistently underperformed relative to the null and baseline models. In contrast, the best-performing representation was obtained with FastICA, which substantially improved silhouette score and modestly improved downstream classification. This result aligns with prior work that uses ICA decomposition to study modular structure in the *E. coli* transcriptome [2], supporting the applicability of ICA-style embeddings. More broadly, this project demonstrates that autoresearch can function as an efficient mechanism for broad experimentation under fixed constraints: it tested a wide range of dimensionality reduction methods, ruled out several unproductive directions, and then concentrated on targeted tuning of the best-performing pipeline.

5 Conclusion

This project demonstrates that autoresearch can efficiently navigate a large model search space while operating under practical runtime constraints. Across 58 experiments, the agent identified FastICA as a strong dimensionality reduction method and incorporated balanced class weights in Logistic Regression to produce a substantial improvement in embedding separability and a modest improvement in classification performance within a reasonable runtime. Evaluation on a held-out test set confirmed the generalization of these gains on unseen data, and that overfitting was not a major concern. At the same time, overall performance remained modest, suggesting that additional improvements will likely require synthesizing domain knowledge from the literature. Overall, these results support autoresearch as a practical tool for efficient and structured experimentation to autonomously optimize a predictive modeling pipeline.

References

[1] A. Karpathy, "autoresearch," GitHub repository. Accessed: May 18, 2026. [Online]. Available: https://github.com/karpathy/autoresearch

[2] A. V. Sastry, Y. Gao, R. Szubin et al., “The Escherichia coli transcriptome mostly consists of independently regulated modules,” *Nature Communications*, vol. 10, no. 1, p. 5536, 2019. doi: 10.1038/s41467-019-13483-w.

[3] M. J. Hawrylycz et al., “An anatomically comprehensive atlas of the adult human brain transcriptome,” *Nature*, vol. 489, no. 7416, pp. 391–399, Sep. 2012, doi: 10.1038/nature11405.

Note on AI Usage

Beyond the explicit applications of the AI agent in autoresearch, AI assistance was also used for initial project brainstorming, structuring and implementing code, writing documentation, formatting figures, and proofreading.